

## EXPERIMENT 12

12

**Study of the Oxidation and Addition Reactions of Carbon Compounds****Aim**

To study the oxidation and addition reactions of carbon compounds.

1. (A) Oxidation of Ethanol
2. (B) Addition reaction of fatty acids

**Requirements**

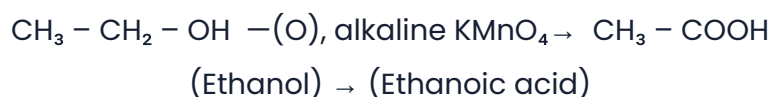
Test tubes, burner, measuring cylinder, dropper, ethanol, dilute solution of sodium carbonate, dilute solution of potassium permanganate, tincture iodine solution, vanaspati ghee, and vegetable oils like groundnut, safflower and sunflower oil.

**(A) Oxidation of Ethanol****Procedure**

1. Take 2-3 ml of ethanol in a test tube, add 5 ml of sodium carbonate solution to it, and warm the mixture.
2. Add a few drops of dilute potassium permanganate solution to the warm mixture while stirring continuously.
3. Observe how the pink colour of the potassium permanganate changes right from when it is added.

**Observations**

1. When a solution of potassium permanganate is added to ethanol, in the beginning the pink colour disappears.
2. When addition is continued further, the pink colour does not vanish but remains as it is.

**Chemical Reaction****Inferences**

1. Potassium permanganate oxidizes ethanol to Ethanoic acid.
2. As potassium permanganate is consumed in this reaction its pink colour vanishes.
3. When the oxidation process is over, the potassium permanganate added afterwards becomes excess and shows its presence with its particular pink colour.

### (B) Addition Reaction of Fatty Acids

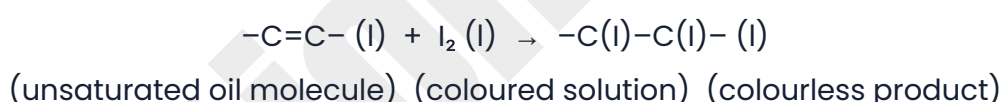
#### Procedure

1. Take 4 ml of oil in a test tube and add 4 drops of tincture iodine to it.
2. Shake the test tube well.
3. Observe the colour change of the tincture iodine.
4. Repeat the same procedure using other oils and vanaspati ghee.

#### Observation Table

| Sr. No. | Oil Used       | Colour Change Observed in the Solution           |
|---------|----------------|--------------------------------------------------|
| 1.      | Groundnut oil  | Orange brown colour of tincture iodine vanishes. |
| 2.      | Safflower oil  | Colour vanishes                                  |
| 3.      | Sunflower oil  | Colour vanishes                                  |
| 4.      | Vanaspati ghee | Colour does not vanish                           |

#### Chemical Reaction



#### Inferences

1. Iodine is consumed due to its addition to fatty acid. Therefore, the coloured solution becomes colourless. But when the same procedure is followed for vanaspati ghee, a similar colour change is not observed. As vanaspati ghee is saturated hydrocarbon, the addition reaction does not occur there.
2. The iodine test indicates the presence of a multiple bond in vegetable oil.

#### Multiple Choice Questions

1.  $-C(=O)-OH$  is called .....

- (A) carboxylic acid group (B) aldehyde group  
(C) ketonic group (D) alcohol group

**Answer: (A) carboxylic acid group**

**2. .... contains alcoholic functional group.**

- (A)  $\text{H}-\text{C}(\text{H})(\text{H})-\text{OH}$  (methanol)
- (B)  $\text{CH}_3-\text{C}(\text{H})(\text{OH})-\text{H}$  (ethanol)
- (C)  $\text{CH}_3-\text{C}(\text{CH}_3)(\text{OH})-\text{CH}_3$  (tertiary butanol)
- (D) all of these

**Answer: (D) all of these**

**3. In the reaction of oxidation of ethanol,  $\text{KMnO}_4$  oxidises ethanol in .....**

- (A) acidic medium (B) alkaline medium
- (C) neutral medium (D) strong acidic medium

**Answer: (B) alkaline medium**

**4. The organic compounds having double or triple bonds in them are termed as .....**

- (A) saturated compounds (B) stable compounds
- (C) volatile compounds (D) unsaturated compounds

**Answer: (D) unsaturated compounds**

**5. Due to ....., vegetable oil is converted into vanaspati ghee.**

- (A) Hydrogenation (B) Chlorination
- (C) Dehydration (D) Polymerization

**Answer: (A) Hydrogenation**

### Remember

Unsaturated compounds undergo addition reaction to form saturated compound.

Unsaturated fats containing double bonds are healthy while saturated fats are harmful to health.

Date: \_\_\_\_\_

Teacher's Signature: \_\_\_\_\_